

Mind the gap: word-embedding and multi-layered literary networks

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Abstract

In recent years, computational literary studies (CLS) have sought to analyse human interactions using social network analysis in prose literature. However, common methods based on co-occurrences often fail to capture the complex, multidimensional, and inter-textual relationships between characters, as described, for instance, by Mikhail Bakhtin. Computational-assisted literary networks might represent the entire character system in a given work, an entirety that human readers usually can't otherwise see, but in a somewhat flat manner. This paper presents an innovative discrepancy-based approach to the uncovering of hidden connections between characters, offering a richer understanding of literary texts. By computing different character relation graphs (such as, co-occurrence, word-embedding-similarity, and annotated-based conversational networks), and taking account of their differences, we can generate an integrated visual model that reveals associations between characters beyond the initial relation graphs—to be then presented to a human reader as a 'teaser' for further reading. The method was tested in the works of Amos Oz, whose novels are characterized by intricate character interactions. Our findings demonstrate that discrepancy networks can identify subtle connections, providing new perspectives on character dynamics and narrative structure. These results are interpreted through a heuristic framework, emphasizing the importance of integrating computational methods with traditional literary analysis. The approach we outline here, enhances the study of individual texts and offers a method for cross-book analysis, contributing to a broader understanding of literary history and theory.

Keywords: computational literary studies; social literary network analysis; word embeddings; digital humanities.

1. Theorizing social literary networks

'When one analyses an individual sentence apart from its context', wrote Mikhail Bakhtin (1986, pp. 99–100) in an unfinished article published only after his death, 'the traces of addressivity and the influence of the anticipated response, dialogical echoes from others' preceding utterances, faint traces of changes of speech subjects that have furrowed the utterance from within—all these are lost, erased because they are all foreign to the sentence as a unit of language'. The reason for this, Bakhtin continued, is that seeing the syntactic boundaries of a sentence as the boundaries of its full meaning as an utterance—that is, as a dialogical act—is to be detached from reality: 'All these phenomena are connected with the whole of the utterance, and when this whole escape the field of vision of the

analyst they cease to exist for him. Herein lies one of the reasons for that narrowness of traditional stylistics ... A stylistic analysis that embraces all aspects of style is possible only as an analysis of the whole utterance, and only in that chain of speech communion of which the utterance is an inseparable link' (Bakhtin 1986). Traces, influences, dialogical echoes: All the things that make human interactions so complex and so difficult to formulate—in real-life communication, as well as in their fictive representation as literature—are also all the things that give Bakhtin's dialogue theory its famously interpretive and philosophical power (Bakhtin 1981).

For about fifteen years, computational literary studies (from now on, 'CLS') have sought to analyse human interactions in prose literature using the most

suitable tool, which is social network analysis (see, e.g. Elson *et al.*, 2010; Moretti 2011; Sack 2013; Prado *et al.*, 2016; Skorinkin 2017; Skorinkin and Fischer 2021). Unlike other CLS attempts, it seems that this one had no clear tradition in pre-digital literary studies; while generations of scholars tried to understand specific relationships between characters, there were relatively few attempts to explore characters' relationships as an entire system (see, e.g. Woloch 2004).¹ Bakhtin himself, while paying attention to the importance of the gallery of characters, of the dialogue as a fundamental generic aspect, and of the tendency of some literary genres to cross hierarchical social boundaries, did not describe the entire character system as such. Nevertheless, the question is whether, and to what extent, existing computational approaches can meet the conceptual challenge his writing demonstrates so clearly. The question is therefore not only—to use but one example—how to determine the centrality of a character (Moretti 2013) when different counts of different elements in the text will yield different results, but how to capture a deeply complex structure without making it too flat; without losing all these traces, influences, dialogical echoes, and more.² As we will see, for understandable practical reasons, computational methods suffer from the same narrowness that prevents us, *human* readers, from seeing the phenomenon in its multidimensional complexity. What binds two characters to one another—the communication that happens between them? They're on stage together, in the same scene? What about their silences or their physical gestures? What about references to them by other characters? What about indirect connections created by analogies or contrasts between seemingly unrelated characters? What about connections mediated by figures *outside* the novel—from other novels, from literary history, from mythology? Furthermore, it should be emphasized that this is not only a big data analysis challenge; even the network representation and analysis of a single novel—or even a short story—may miss the complexity of the relationships between characters due to their multi-layered and complex nature.

It is true that, in at least one aspect, social literary networks do see more than human readers can: They view the work—or, the communication systems within it—as a *whole*; they capture that 'inseparable link' that sometimes gets lost through reading. At least in that sense, they see *everything*: 'When we watch a play, we are always in the present: what is on stage, is; and then it disappears. Here [= in the network graph], nothing ever disappears. What is done cannot be undone [...]. The past becomes past, yes, but it never disappears from our perception of the plot' (Moretti 2011). The ability to see all the connections in a work at once, from a bird's-eye view, regardless of whether they

persist or have ended, is certainly more than one can hope to achieve while reading. The benefits of this are indeed numerous. However, so long as the parameters on which the network is based can be extracted and measured easily (in a process that can, in theory, also be done manually), the network will be limited in one way or another. If the computational added value consists mainly in applying the same parameters to a larger corpus or finding the most efficient automated method, the results—good as they might be—will not always help us ask more fundamentally what social literary networks *are*.

Like many other computational text modelling, from simple word clouds or literary maps to complicated topic modelling or stylometric clustering, typical graphical representations of social literary networks are an excellent way of looking at the whole, but they come at a price: They push us to see the whole as somewhat flat and limited, both quantitatively and qualitatively, within the parameters that distinguish its different aspects from one another. This is not to say that they are not useful; in fact, they have extraordinary interpretive value. But, as common and fruitful a CLS method as it is, it seems that the idea of an 'undogmatic reading' as a vision for reflective computational research, as suggested by Jan Christoph Meister (2014),³ seems not to have been achieved yet. This limitation characterizes all algorithmic representations of literary networks, including those we will propose later. It is inherent to the method. We believe, however, that it can be at least partially overcome.

2. From co-occurrence networks to word-embedding networks

Traditionally, the most common literary network is based on co-occurrences: If characters appear in the same scene, they are linked. A powerful example of this can be found in the DRACOR project (Fischer *et al.*, 2019),⁴ which displays the network of each play in its corpus. As all the plays are well TEI-encoded, and as all of them are plays and not novels, it is quite easy to automatically divide the text into scenes or acts and identify the characters in each. For example, Figs 1 and 2 present Lessing's *Nathan der Weise* and Schiller's *Die Räuber*, written in almost the same year (1779 and 1781, respectively).

Clearly, here we have two different networks, with two different poetics. Both have a central area with just a few characters who have strong connections. However, while Lessing's play has a small number of minor characters—some of them are connected to the main characters only by loose links—Schiller's is structured by a highly complex dense network of a relatively large number of minor characters. This tool is

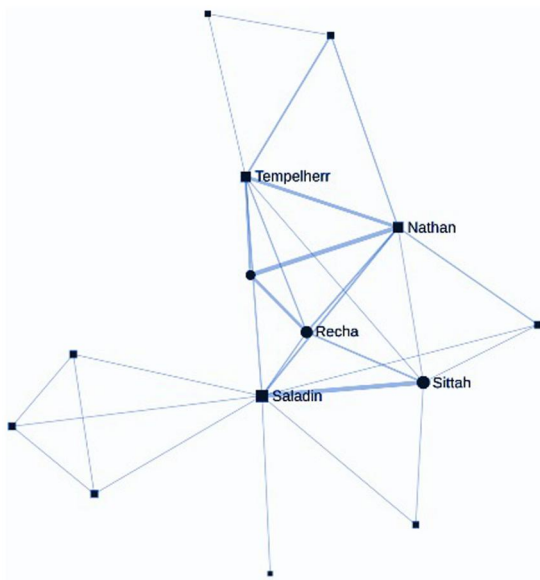


Figure 1. DRACOR's network [Fischer *et al.* (2019)] of Gotthold Ephraim Lessing's *Nathan der Weise*. See: <https://dracor.org/ger/lessing-nathan-der-weise> [29.05.2024].

designed for distant reading, almost in the original sense of the term: by the simple press of a button, it allows users to obtain networks for plays they've never read before and to learn quite a bit about them. This result does not necessarily meet the fundamental conceptual challenge described above, since being in the same scene is only one expression of the relationship between characters. As important as it is, it tells us very little about the emotional relationship between the two characters, nor about their role in the deep structure of the plot. For example, close characters whose differences dramatize an ideological contrast at the core of the narrative will be represented in the network like other close characters who do not dramatize such a contrast. Or an even simpler example: Two characters that appear together in many scenes but never exchange a word, will be represented in the network like another pair that never appear together but communicate profusely via a third character. These are perhaps two extreme cases, but they do illustrate the deficiency of such an approach.

Using a similar database of novels rather than plays would have been more complicated: in the absence of scenes with clear boundaries, it is more difficult to extract connections between characters. But the common method and its benefits are similar: If characters appear in the same sentence, paragraph, or text window of a given length—which replace the encoded fixed scenes in plays—they are represented as linked, so that the researcher is able to see at least one aspect of the literary network in its entirety. Other methods are

focused on different aspects and based, for example, on counting instances of direct communication between the characters—what Elson *et al.* (2010) and others have called 'conversational networks'.⁵ But these are, as Skorinkin (2017) showed, more difficult to extract automatically.

The question remains, however: how to move from these methods—each highlighting one different aspect of the network—to a computational technique that will lead to a deeper conceptualization of literary networks, getting as close to their *multidimensional* essence as possible, and a better understanding of their place in literary theory and history.

The research question that we study is: *Can discrepancies between different models of literary character networks function as productive interpretive triggers for close reading.*

Our proposal is built upon three principles: first, drawing on existing network types that have been previously studied, such as co-occurrence and semantic relationships, with a focus on text-structure connections; second, developing a useful way to use different networks together, in a comparative manner; and third, enhancing the reader's role as a subject who deciphers these networks, analyses their achievements and limitations, and gives them meaning.

With Bakhtin's dialogic theory in mind, the first step will be to expand the boundaries of the units of analysis. In other words, to see every character, and every utterance made by that character, in the light of (at least) the entire text. Thus, one potential direction—though certainly not the only one—will be to build a network based on word-embedding algorithms, such as word2vec or BERT, since these can identify hidden connections between words that may be far apart in the text or may even be taken from different texts. These algorithms can identify a connection between words like 'milk' and 'yoghurt', 'Paris' and 'London', 'king' and 'queen'—based on the hypothesis that 'words that appear in the same context tend to have similar meanings' (Harris 1954), when 'similar meaning' is measured by cosine similarity, and the question is whether such algorithms are capable of recognizing hidden links between 'remote' literary characters.

The initial answer is that, to a certain extent, they can. The outcome does not fundamentally differ from conventional methodologies in the sense that it, too, points out only one aspect of the network, although an aspect of a different type. It is a graph of what might be called—though not to the fullest sense of the term—a *semantic* network of literary characters. While it is not always clear what kind of 'semantics' connects two characters, and while the co-occurrences of characters undoubtedly create a bias in the same text window, it nevertheless succeeds in pointing out hidden

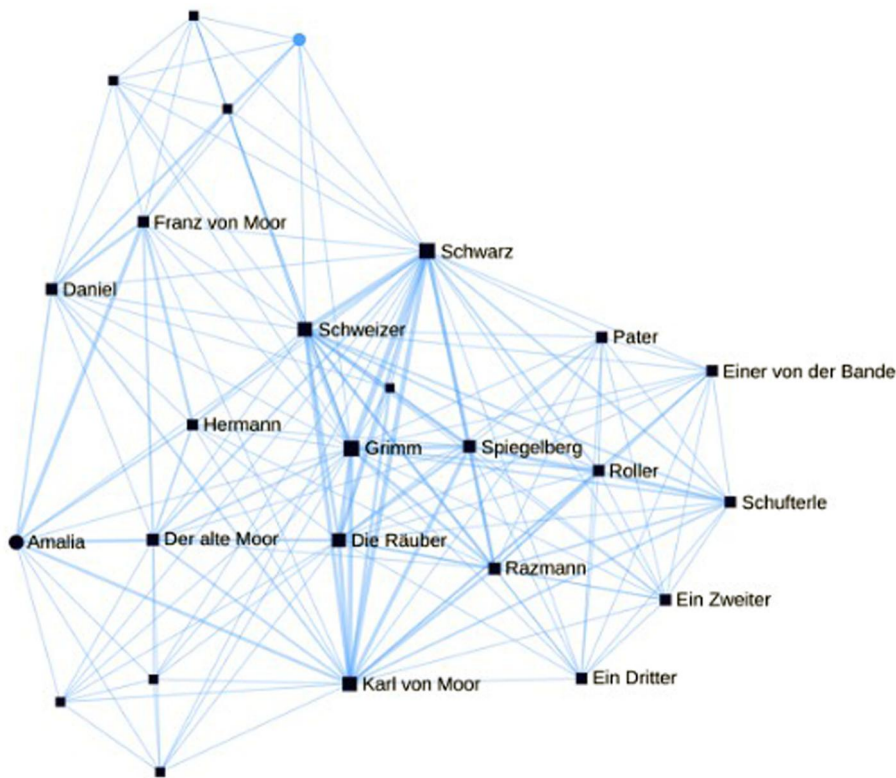


Figure 2. DRACOR's network of Friedrich Schiller's *Die Räuber*. Conceptually, our approach here follows the principled approach of Jan Christoph Meister, as described by Flüh et al. (2021). While Meister applied his approach to text annotation, we apply it to text network modelling.

connections between seemingly unrelated characters. However, as we will see, the results require not only interpretation but also reframing.

This direction was partially taken, for example, in 'Learning and Evaluating Character Representations in Novels' (Inoue et al., 2022), which summarizes a comprehensive attempt to compute a numerical representation of characters. The latter paper's primary objective is to create high-quality distributed representations of characters that capture both local and global contextual information from novels. To that end, a method is proposed in that paper for learning character embeddings using document-level information, specifically graph neural network-based embeddings and low-dimensional embeddings based on character occurrence patterns. It combines these methods with text-based embeddings to improve character representation, and evaluates their effectiveness using a new benchmark suite of twelve tasks. While the authors of the paper demonstrate the usefulness of revealing patterns that the human reader can also reveal (in some cases, even this is not achieved well, especially in context-level tasks), in the end, no surprising or hidden patterns were discovered. No piece of

information will force the reader or the researcher to re-evaluate the book he or she just read. The purpose of the study is not to provide new answers to literary studies inquiries but to develop a better computational framework for literary data-processing. The question of whether such a method can be integrated into traditional literary research remains open.

3. Modelling discrepancies: back to the reader

Our own framework, in contrast, aims to work with precisely these literary questions in mind and to find those spots exactly—or, at the very least, to draw the reader's attention to potentially interesting spots. What matters most to us is not necessarily the specific methodology of a semantic network—which is only an example, and, like any other network, will tend to highlight some aspects while obscuring others. Rather, it is the very attempt to expand the possibilities of network representation to connect characters' relations to different dimensions of the text.

Admittedly, the challenge of finding those spots by developing a new framework is doubled since the

framework's designers (ourselves, the researchers, who are also readers) may not be aware of those spots to begin with. Thus, a logical next step would be to *compare* network dimensions rather than focus on only one. But how can literary network comparisons be made? Our modest starting point is to consider pairs of characters instead of a single character as a basic entity under study. This technical decision allows us to compare the different networks, and more specifically, to find *discrepancies* between the representation of a pair in one lens (say, co-occurrence) versus their representation in another (say, word embedding). These discrepancies, or gaps, will be our hidden way in. As we shall demonstrate, this approach has the potential to provide a more nuanced view of character interactions and to highlight the benefit of combining multiple computational methods to enhance literary analysis.

Why would one expect that looking at pairs and gaps can provide us with such an opening? Let's do a simple mathematical computation to illustrate it. Suppose we have n main characters in the novel, and suppose that there are k lenses (social networks in our case) that the reader uses to understand these characters and the relationships between them (co-appearance is one such semantic lens, word context is another). The number of unordered pairs of characters is $\binom{n}{2}$; similarly, there are $\binom{k}{2}$ pairs of semantic lenses where gaps can arise. All in all, the reader is supposed to follow $\binom{n}{2} \cdot \binom{k}{2}$ 'routes'. Plugging in $n = 5, k = 3$ (a modest estimation for n, k), we get 30. No human brain can track 30 routes sufficiently well to come up with meaningful conclusions. A machine, however, can do it easily.

The next question is, how to demonstrate the usefulness of our approach. When a new approach is tested, the standard way in NLP/Computer Science is to measure its performance on a batch of downstream tasks. For example, here are two tasks from (Inoue *et al.*, 2022). Task 1: Given a sentence S with a blank (e.g. '___ stood up and tried to keep her eyes open while Mrs. Medlock collected her parcels') from book B and four candidate characters from book B , choose the character C that best fits into the blank. Task 2: Given a quote Q (e.g. 'Well, it was this way. I was leaning on the stile...') from book B and four candidate characters from book B , choose the character that spoke these words.

Generating such tasks depends on human labour, annotation, and orchestration. Yet this contradicts the very logic of the trapdoor paradigm. Once again, we find ourselves in a dilemma: if we ask readers to label the data and researchers to construct the tasks, we end up contributing more to the computational product than to the reader's.⁶

Thus, it seems that the best way to evaluate our framework would be to do so in an *unsupervised* manner and in reverse order. Rather than a labelling procedure that precedes the pipeline and is independent of it, we suggest having readers perform a close-reading interpretation of the insights which the framework offers in the *aftermath* of their reading of the book. The computational output, which we call 'a teaser', is given to the literary scholar, or reader, asking them: Does this make sense to you? And please, 'mind the gap': What can *you* learn from the *discrepancies* between the two measurements? In other words, instead of seeing the computational product as the heart of our research and its end result, we deliberately offer it as the beginning of a process that is essentially human (Marienberg-Milikowsky *et al.* 2022). Rather than producing the best graph from a purely digital point of view—a graph that can be applied to as many texts as possible automatically to extract reliable network information—our goal is to create the best graph from a traditional humanistic point of view: a graph that will serve as food for thought for literary researchers. It should be noted, of course, that this graph also diminishes the richness and complexity of the relationships between literary characters; however, the fact that the representations of these relationships *differ*, as emphasized by the graph itself, illustrates the principle of partial analysis and interpretation, and therefore provokes the reader's reflection.

It is no coincidence that the reader's response theory, which flourished in literary studies during the seventh decade of the last century, has a part to play in this method.⁷ Consequently, this method goes beyond the concrete embodiment of the written words by recognizing that different readers can interpret texts differently and that meaning depends on interpretive acts. Additionally, these method's 'teasers' can also serve as food for thought for any reader, as part of a so-called 'public humanities' or 'citizen-science' project, which recently formed part of several CLS projects (see, e.g. Marienberg-Milikowsky and Dekel 2021; Schumacher *et al.*, 2023).⁸ In such a case, the teasers will be made publicly available, and readers—professional or not—will have the opportunity to interpret them in several ways, from free text response to a structured annotation. Figure 3 illustrates the difference between the task-oriented pipeline and the human-oriented one.

In what follows, after describing our test-case corpus and a few technical issues, we will discuss two potential applications of this method: creating a multi-layered graph as a teaser for re-reading one book; and creating multi-layered aids as teasers for re-reading across books.

Up to this point, we have focused on how our method differs from common network analysis approaches. To

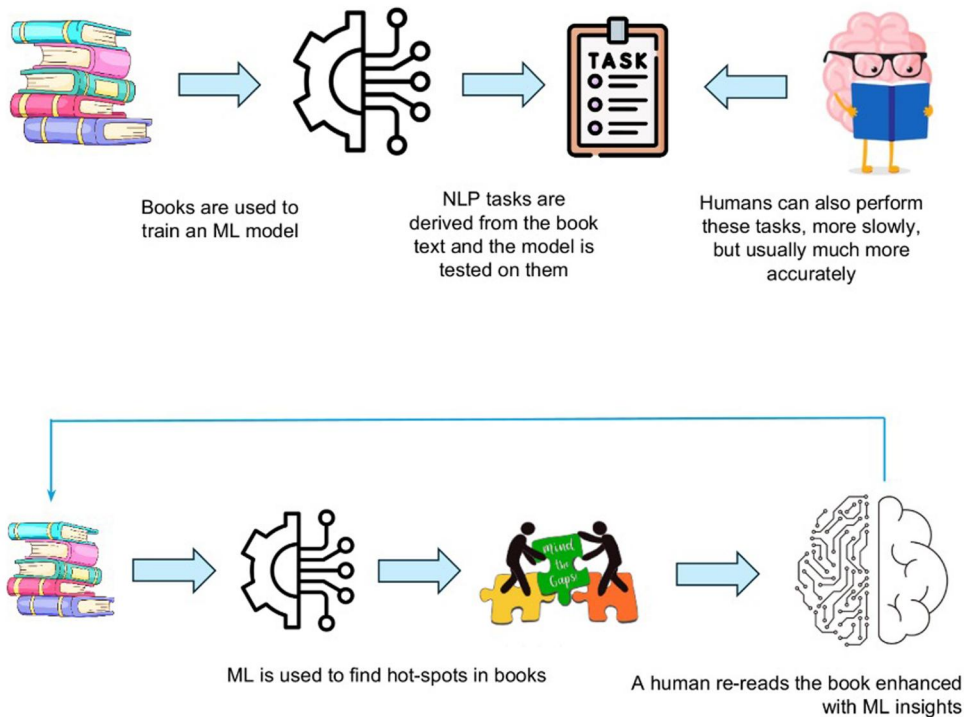


Figure 3. Top—computer-oriented pipeline; bottom—reader-oriented pipeline.

further place our contribution, we now compare it to a related but distinct line of work: graph distance and network comparison methods. While our method does not compute a formal graph distance, it shares conceptual similarities with the broader network comparison task, as Tantardini *et al.* (2019) surveyed. However, rather than quantifying an overall distance between networks, we focus on identifying *pairwise discrepancies*—instances where two characters are strongly connected in one network but not in another. This approach aligns more closely with the spirit of localized difference detection than with global network similarity. It diverges from typical methods like NetLSD or Portrait Divergence, which aim to produce a single scalar measure summarizing global structural differences. Instead, our goal is to direct attention to specific, semantically meaningful gaps—teasers for interpretive reading—rather than to quantify overall similarity.

Furthermore, our goal is not just to detect differences between networks but to interpret them—to relate discrepancies to literary meaning. This requires metrics that are transparent and easily understood. Methods like NetLSD, which rely on spectral signatures from heat kernel traces, compress the network structure in ways that are hard to relate to individual characters

or scenes. In contrast, our edge-wise comparisons—such as contrasting word2vec similarity with co-occurrence—highlight intuitively meaningful and gaps that are readily interpretable by literary scholars.

4. An initial test case and technical considerations

As Hebrew literature scholars, we decided to focus on the work of Amos Oz, who wrote about 25 novels over nearly 50 years—a sizeable corpus, but not too large, and one that reflects a relatively long artistic development process. Besides having Oz's archive at our disposal,⁹ we chose him because the characters' interactions are central to his writing and the scholarship dedicated to it. No less importantly, we chose him because of his status as one of the most translated writers in Hebrew, allowing us to process his English translations using useful packages such as BookNLP (Bamman 2021), which go far beyond what is currently available in Hebrew.¹⁰

The protocol that we have used for each translated novel so far has been the same: after pre-processing the text to prepare it for computational analysis, we (1) automatically created a co-occurrence network; (2) automatically created a word2vec distributional

similarity network; (3) manually created a conversational network; and (4) calculate the discrepancies between every pair of networks, and evaluate them from both a computational and a non-computational point of view. Table 1 above summarizes these three methods and their parameters.

While the idea behind the whole protocol is as described above, each one of these stages raised some technical questions we had to address over time, such as how to find the right window size for stage (1), whether to train the model in stage (2) from scratch, and how to create a conversational network manually (and also: how to define a conversation) in stage (3), or even what to do with first-person narrator novels in the entire process (in brief: we decided to avoid them due to the difficulties they pose for the automatic procedures).

Here are a few clarifications. First, the main hyperparameter used for counting co-occurrences of pairs of characters is the window size (N), on which we check whether the pair appears together or not. The procedure is simple: We iterate over all appearances of each character and count co-occurrences with all other characters that appear within the range of N tokens from the character's name token. This way, we ensure that each co-occurrence will be counted only once.

Second, we decided to train a word2vec using CBOW methodology from scratch because a pre-trained model would not contain the entities' names as part of its vocabulary. Another issue is that, even if the names happened to appear in the pre-trained model, they could all be embedded in the same geometric area of 'names' without any nuanced differentiation, certainly not one derived from a specific book.

Third, while the use of word2vec may appear outdated, given the advent of contextual models like BERT and other transformer-based architectures, our choice was guided by the specific requirements of our task. Contextual embeddings such as those produced by BERT generate a different vector for each token occurrence, meaning that for each character name, we obtain a list of context-specific vectors rather than a single unified representation. To compare two

characters, we would then need to reduce these multiple vectors to a single value, typically by averaging them, which effectively collapses the contextual richness into a static form. This reduction process ends up being similar to the output of word2vec, which directly provides a single, global representation per token. Indeed, prior work has shown that averaged BERT embeddings tend to approximate static embeddings like word2vec, particularly when used outside of sentence-level tasks (e.g. Ethayarajh 2019). Having said that, we could have used BERT instead of word2vec: in the few tests that we ran, we noticed that the results didn't change significantly.

To determine the optimal window size for word2vec in modelling character relationships, we evaluated the distribution of cosine similarities between character pairs under different window sizes (In Fig. 4, we show some of our experiments for different word2vec window size choices 5, 10, and 20). As shown in the box-plot in Fig. 4, for *A Perfect Peace*, a window size of 5 yielded the largest variance in cosine similarity scores, indicating high differentiation in how character relationships are encoded. While variance is desirable—suggesting that meaningful semantic distinctions are being captured—excessive variance may reflect instability or noise in the embeddings. Conversely, a window size of 20 resulted in very low variance, implying homogeneity in vector directions and possibly a loss of expressive power. We therefore selected a window size of 15 as a balanced compromise: it maintains substantial variance (indicating that the model detects nuanced differences) while avoiding the potential over-randomness seen with a very small window and the flattening effect seen with a very large one. We should also note that the word2vec windows were restricted to stay within chapter boundaries, so some edge windows are smaller. Also, we used the same window size for the automatic co-occurrence count.

Fourth, regarding the manual creation of conversational networks: We decided to base this part of the research on the simple-to-use EZlinaviz application (Fischer and Milling 2017),¹¹ created as part of the DRACOR project, to keep the level of technical

Table 1. A summary of the three networks used in this paper.

Network name	Algorithm used	Parameters/output
Co-occurrence	Sliding window token proximity	Window size = 15 tokens (restricted within chapters); counts co-occurrences of character pairs, respecting chapter boundaries.
Word2vec (distributional similarity)	Word2vec (CBOW, trained from scratch)	vec_size = 200, context_win = 15; outputs cosine similarity between character embeddings; window respects chapter boundaries
Manual conversational	Human annotation via EZlinaviz	Verbal interactions manually annotated; edge weight = number of verbal exchanges

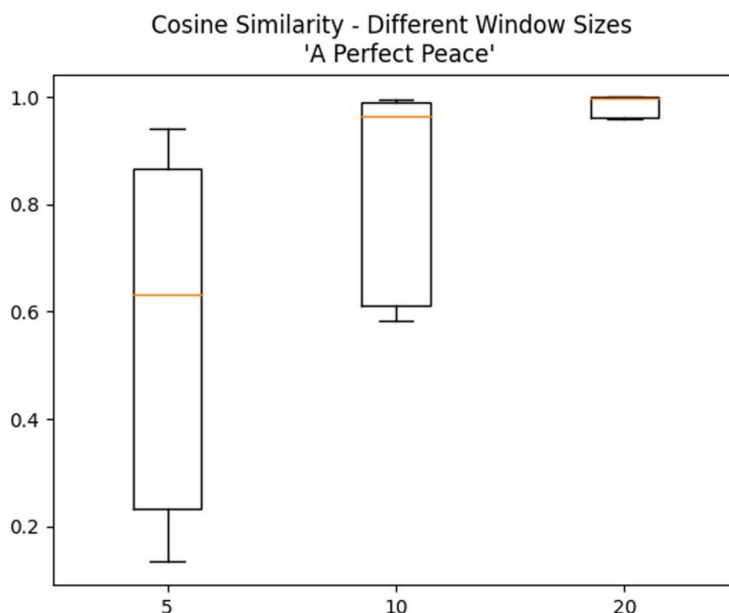


Figure 4. A box plot of the cosine similarity for different window sizes in Amos Oz's book *A Perfect Peace*. The plot is computed over pairs of characters (y-axis) against the window size (x-axis). We see that the larger the window, the smaller the variance, which makes sense as the characters are 'swollen' by the text.

difficulty to a minimum and help our reader (a traditional literary scholar by training, with basic knowledge of DH and CLS) to build a network without having to learn advanced software.¹² With this application, the reader can model relationships between characters intuitively using text that has not been coded or digitized. Thus, instead of defining a relationship based on the division of a text into scenes, as in the original implementation of the tool in the original architecture of DRACOR, we enabled readers to define a relationship through the verbal communication between two characters. Relationship strength is measured by how many verbal communication events occur between the characters. In addition, unlike the co-occurrences model, the segments were not equal in the number of tokens but were determined by our plot interpretation.

Finally, we would like to discuss the technicalities of how exactly the gaps are computed. Following steps (1)–(3), the edges of the input graphs are weighted with the relevant metrics (e.g. co-appearance count or word2vec cosine similarity). A naive attempt to compute the difference would be to take the edge-weight difference in the two graphs. For example, given two characters a and b , look at the weight w_{ab} in G_1 , and w'_{ab} in G_2 and let set $|w_{ab} - w'_{ab}|$ be the new edge weight in the difference graph (the difference between the weights in absolute value). The problem with this approach is that the units of w and w' need not match. Co-sine similarity is a number in $[-1, 1]$, while co-

appearance can be in the hundreds. Normalization can solve this problem, but also binning. The latter solution reduces the number of weight options and allows for easier interpretation. To perform binning, we use the k -means clustering algorithm. Specifically, we take the list of edge-weights in graph G_1 and cluster using k -means. We do the same for G_2 . Now, each edge can be re-weighted with the cluster number. Since we are clustering positive numbers of the real line, the lower the cluster number, the higher the original weight (assuming we start labelling from left to right on the number axis). We now generate the difference graph G_{diff} , which has the same vertex set as G_1 and G_2 , and each edge is weighted according to the difference in cluster number. The graph in Fig. 6 is such an example.

One technical note on how to choose k . We order the edge weights from small to large and plot (i, w_i) , where i is the pair number in the ordering, and w_i is its weight. We look at the graph, and it turns out that a step structure is obtained. Every 'step' intuitively corresponds to a new cluster. We count how many steps there are, and that's our k .

5. Initial results: multi-layered networks as a teaser for re-reading

We focused on Amos Oz's novel 'To Know a Woman' (1992 [1989]), in which the protagonist, a former secret service agent named Yoel Ravid, tries to use his professional skills to better understand his

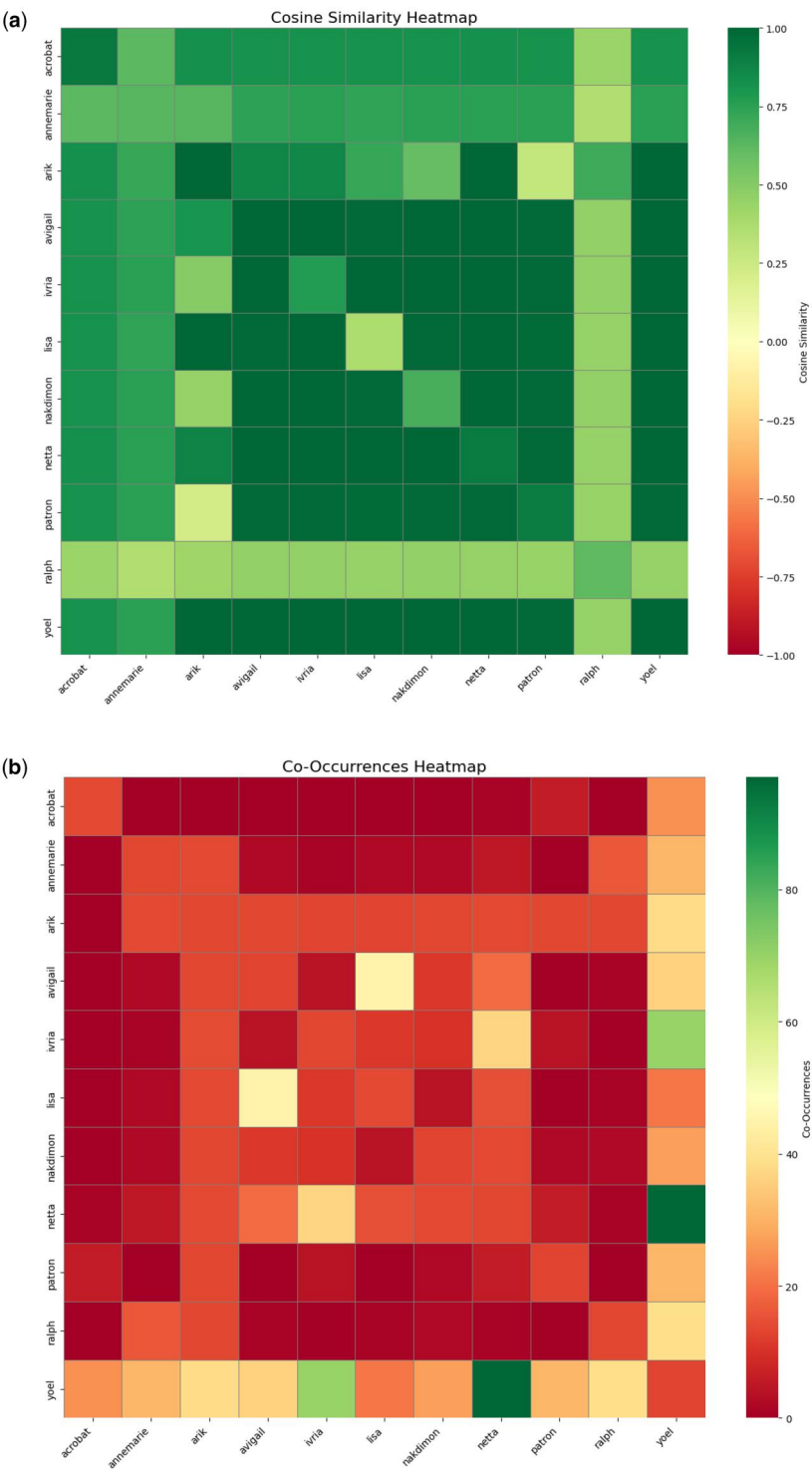


Figure 5. Two networks of 'To Know a Woman'. (a) word2vec network; (b) co-occurrences network. The networks are described using a heat map of the weighted adjacency matrix.

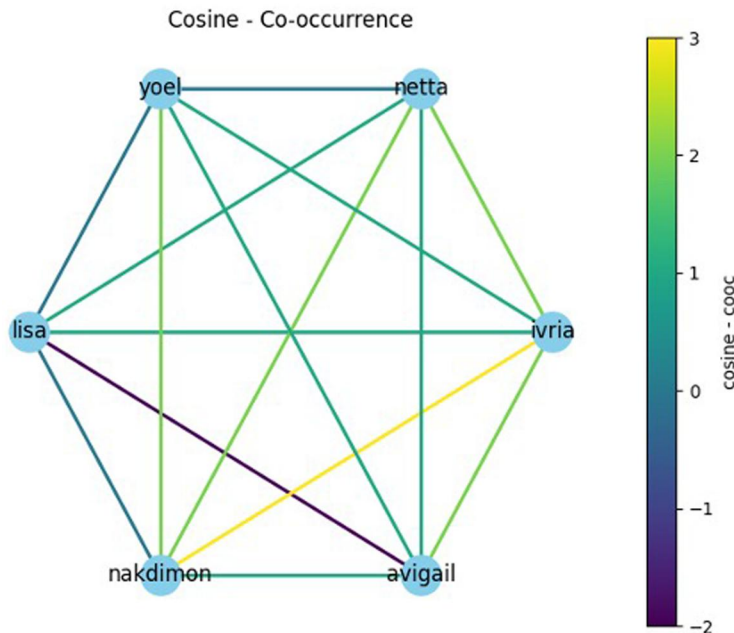


Figure 6. A gap network (cosine minus co-occurrences).

relationships with his daughter, mother, and mother-in-law; in many ways, a typical novel of Oz, written in the middle of his career—and one that manifests the complexity of family relationships. Figure 5 features two different graphs of ‘To Know a Woman’. The left one is a word2vec network (training a W2V model from scratch with the following hyperparameters: `vec_size = 200`, `context_win = 15`), and the right one is a co-occurrences network (counting how many co-occurrences each pair of characters has in a predefined sliding window of size 15).

Even without going into detail, it is plausible that each network tells a slightly different story about the same novel; thus, they are a good fit for discrepancy mining. The most interesting gap for a reanalysis of the novel is the change in the place of secondary characters: In the co-occurrence network, the main characters (the core family: Yoel, Netta, and Ivriya) make up the innermost circle, while in the word2vec distributional similarly network only Netta remains at the centre, and minor characters (such as Nakdimon—Ivriya’s brother—and Dubi, Netta’s partner) gain centrality. Considering that the novel is mainly told by an external narrator adopting the protagonist’s perspective by using, for example, a free indirect speech technique, the word2vec network allows the book to be read more polyphonically. From a narratological angle, this is a significant difference, and not just for the discussion of this particular novel.

Based on different methods, Moretti (2013) called a similar experiment ‘concept against concept’, and

emphasized that ‘it’s not that one is right and the other is wrong; rather, they capture different features of dramatic networks’.¹³ However, while Moretti compares them manually, we suggest doing it computationally, that is, to put one on top of the other and calculate the gaps between them.

Looking at Fig. 6, the nearer the edge’s colour to yellow or dark purple, the stronger the ‘disagreement’, or discrepancy, between the two models. In the graph, we see that the value of discrepancy is positive for most pairs. That is, they have stronger co-sine similarity than co-appearance. The purple pairs will have the opposite relation. They appear much more than their semantic connection would seem to warrant.

Table 2 table summarizes our findings for ‘To Know a Woman’.

In addition to the mathematical evaluation of it, and as an interpretive teaser—as food for thought—the meaning of such a method has to be interpreted, first and foremost, by human reading. And here some interesting clues come to mind—again, without going into too much detail. We can see, for instance, that the triangle between the main characters (Yoel, Netta, and Ivria) is relatively green, close to blue (namely, a small discrepancy). As the main characters, they interact a lot, they have a lot in common, and, not surprisingly, they have a strong semantic relationship. What appears to be more intriguing is the connection between the main and secondary characters. As an example, Nakdimon, Ivria’s brother, appears only a few

Table 2. Summary of ‘hot spot’ pairs, pairs where the semantic cluster is larger than the co-occurrence cluster.

Pair	Hot spot	Evidence in the graph	Teaser?
Yoel–Netta	No	Green/Blue edge (low discrepancy)	No. Strong interaction and semantic bond; central characters with mutual presence and thematic overlap.
Yoel–Ivria	No	Green/Blue edge	No. Similar to above—close interaction and aligned semantic context; both are core to the narrative.
Netta–Ivria	No	Green/Blue edge	No. Core character triangle; consistent presence and shared themes yield low discrepancy.
Yoel–Nakdimon	Yes	Green/Yellow edge (co-sine > co-occurrence)	Yes. Despite few co-occurrences, high semantic similarity suggests hidden psychological or thematic resonance.
Netta–Nakdimon	Yes	Green/Yellow edge (strong discrepancy favouring co-sine sim)	Yes. A teasing link; possibly reflective of Nakdimon’s ability to voice hidden tensions, or embody Yoel/Netta’s desires.
Lisa–Avigial	Yes	Purple edge (co-occurrence > co-sine similarity)	Yes. High co-occurrence due to physical proximity and routine, but low semantic overlap—reflects disagreement.

‘Hot spot’ pairs are also listed for comparison. The last column reveals the teaser (or triviality).

times in the novel but seems to have a strong connection with the main characters. This can be seen in the green-yellow and yellow lines between them. In the context of the entire plot, the semantic connection between the siblings is predictable. But what does a simple non-intellectual farmer like Nakdimon have to do with Yoel and Netta, his urban, gentle and well-educated brother-in-law and niece? This is an example of a teaser, a stimulating connection for the human reader. We can suggest several possible interpretations here: It might reflect the hidden desire of Yoel to be more straightforward, more simple, a desire that is embodied, from a structural point of view, in Nakdimon; or maybe it reflects Nakdimon’s tendency—and ability—to voice out loud painful family issues that Netta and Yoel avoid discussing?

An interesting hot spot opposed to this is the connection between Lisa and Avigial, the mothers of Yoel and Ivria (respectively). The purple line between them indicates that they co-occur many times while having a weak semantic connection. They are close friends who moved in with Yoel and Netta after Ivria died. Interestingly, they are often called ‘the grandmothers’ by other characters in the novel because they do almost everything together. On the other hand, while they take care of each other, and are synergistic in their movements and thoughts, they do not agree on the right way to handle the family’s situation. Table 2 summarizes the pairs we discussed, and the type of gap observed.

The same approach can be used to offer the human reader a comparison between each one of the two computational models and the manual one, as shown in Figs 7 and 8.

It appears that the map comparing the manual networks and the cosine similarity network is somewhat similar to the first map comparing the two computational models. This is perhaps due to the partially similar logic of the manual and co-occurrence webs (a similarity that will be ranked high in some novels and low in others). But two major differences stand out, summarized in Table 3, and both are related to mothers and their children: Yoel’s connection with Lisa is purple (compared to the green connection in the previous network); that is, the gap between the two has decreased.

The opposite occurs for Netta’s connection with Ivria, where a green edge is now yellow. For Oz scholars, this is an important hint since motherhood is one of the major themes in his entire work (Schwartz 2020). As many scholars have noted, Oz’s complicated relationship with his own mother (who committed suicide when he was twelve years old, as described in his autobiography; *A Tale of Love and Darkness*) has influenced the depiction of mothers in his novels as detached, cold, and unloving (see, e.g. Abramovich 2009). The question of how concretely to work with this pointer and how to take it into consideration while rereading the novel is beyond our focus here.

Another instructive visualization is the Spearman correlation matrix of the three modelling methods presented in Fig. 9 (computed using the raw values, no clusters, as correlations are computed in a normalized way by definition).

The correlation matrix in Fig. 9 is generated from ‘To Know a Woman’ and offers an interesting point of view on the relation between the different modelling methods. Manual-based conversational networks and

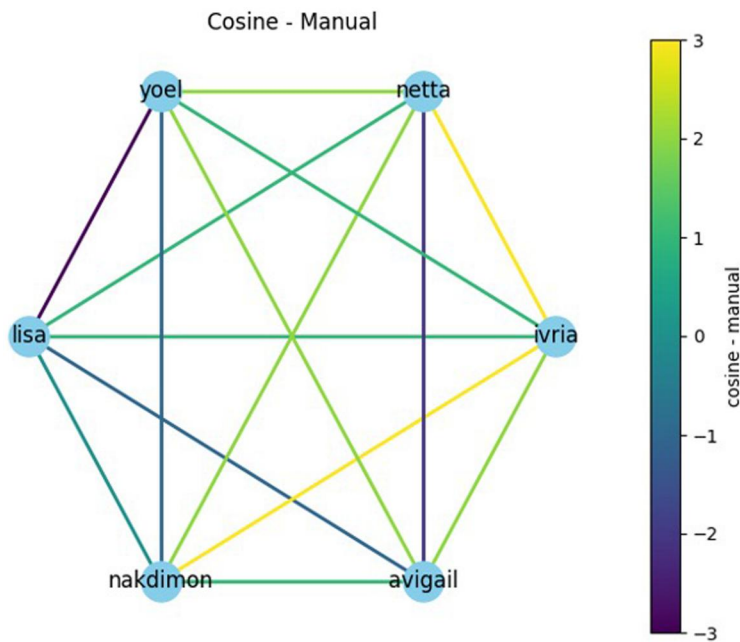


Figure 7. A Gap network (cosine minus manual co-occurrences counting).

automatic co-occurrence networks have a moderate to high correlation, which makes sense. An interesting point is that cosine similarity correlates more with manual counting than automatic (0.4 with manual counting versus 0.51 with co-occurrence).

6. Cross-book readings

Obviously, our method, which can be regarded as an undogmatic heuristic method,¹⁴ only applies in the context of close reading, for better or worse. This method is excellent for analysing one novel in depth, several novels in depth, relationships within a well-defined small corpus, or even as a basis for broader conceptual abstractions. When wanting to analyse large systems, however, in order to better understand cross-book connections and relationships, it becomes less useful.

In this case, we propose first to look at cross-book analysis based on one model—the only model of the three described above that can fulfil this task—the word-embedding model. Here, for example, is a calculation of word2vec-based similarities between characters of five different novels by Amos Oz:

In Fig. 10, we see the characters' names in different colours. Each colour represents the book in which the character appears (these data are not a network but a two-dimensional PCA projection of 200-long word2vec vectors of the characters' names). Characters with

a higher cosine similarity are closer to one another. Unsurprisingly, characters in one novel tend to be close to one another, but this is not always the case. Sometimes a character from one novel may be more semantically related to a character from another.

Again, this finding does not stand on its own. It requires the interpretation and explanation of an experienced researcher familiar with all the novels. But it does bring us a little closer to the vision described earlier. To improve readers' understanding of character relationships across books, we have produced tables detailing the degree of closeness between each pair of characters. Through this method, readers can transform distant reading from viewing the different books as a whole, into distant reading that focuses on the interrelationships between the characters.

Next, we shall add another layer to this visualization to test our gaps-oriented conceptual approach, not only with regard to a single-book analysis but also to cross-book analysis. Obviously, this second map cannot be based on co-occurrences or manual networks; instead, we suggest comparing a word2vec cosine similarity graph with a graph where edge weights represent the differences in publication years of the respective books in which the characters appear.

In order to be consistent, we used `1/years_gap` so that higher values would reflect similarity or closeness.

In the resulting difference graph, shown in Fig. 11, we use node colours to enhance the interpretability of

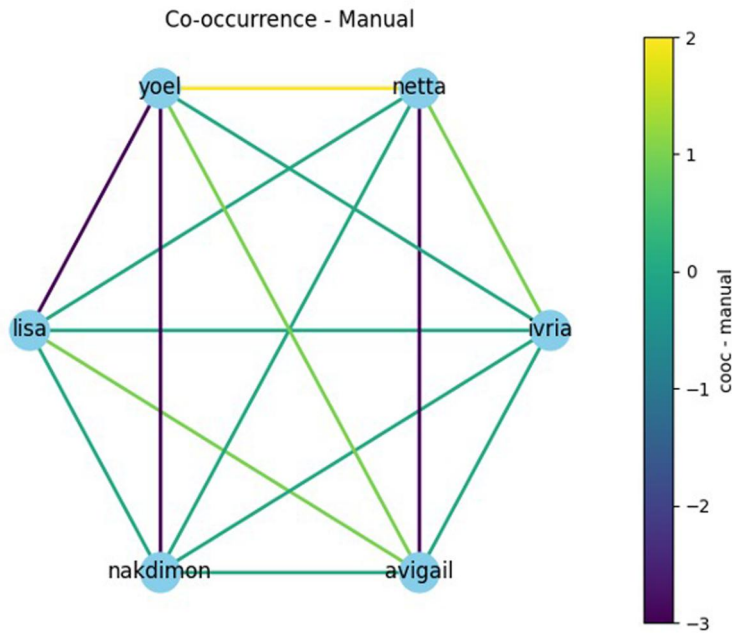


Figure 8. A gap network (automatic co-occurrences counting minus manual co-occurrences counting).

Table 3. Two changes when using manual networks versus automatic co-occurrence; the co-occurrence measure (manual or automatic) is compared to cosine.

Pair	Change in edge (manual versus auto)	Discrepancy direction	Findings
Yoel–Lisa	Green → Purple	Discrepancy decreased	Manual annotations reflect a weaker tie, aligning more closely with semantic similarity. Manual view highlights stronger co-occurrence than semantic similarity would suggest.
Netta–Ivria	Green → Yellow	Discrepancy increased	

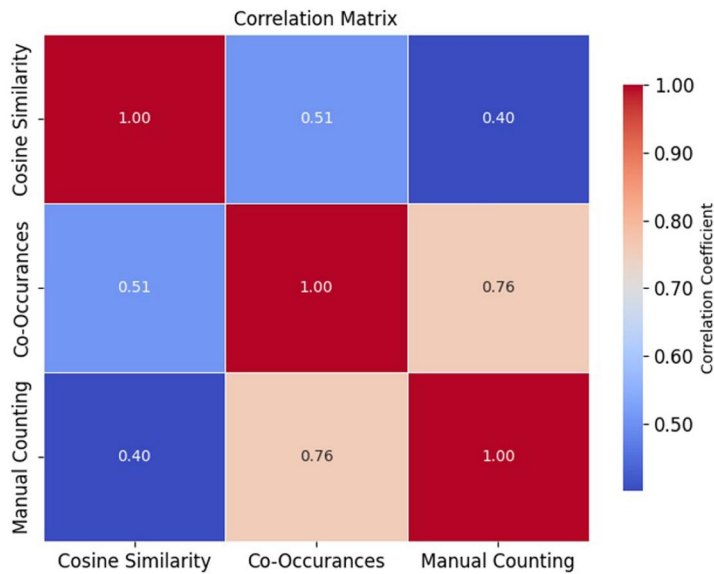


Figure 9. Correlation matrix of ‘To Know a Woman’ (describing the correlation between the 3 models).

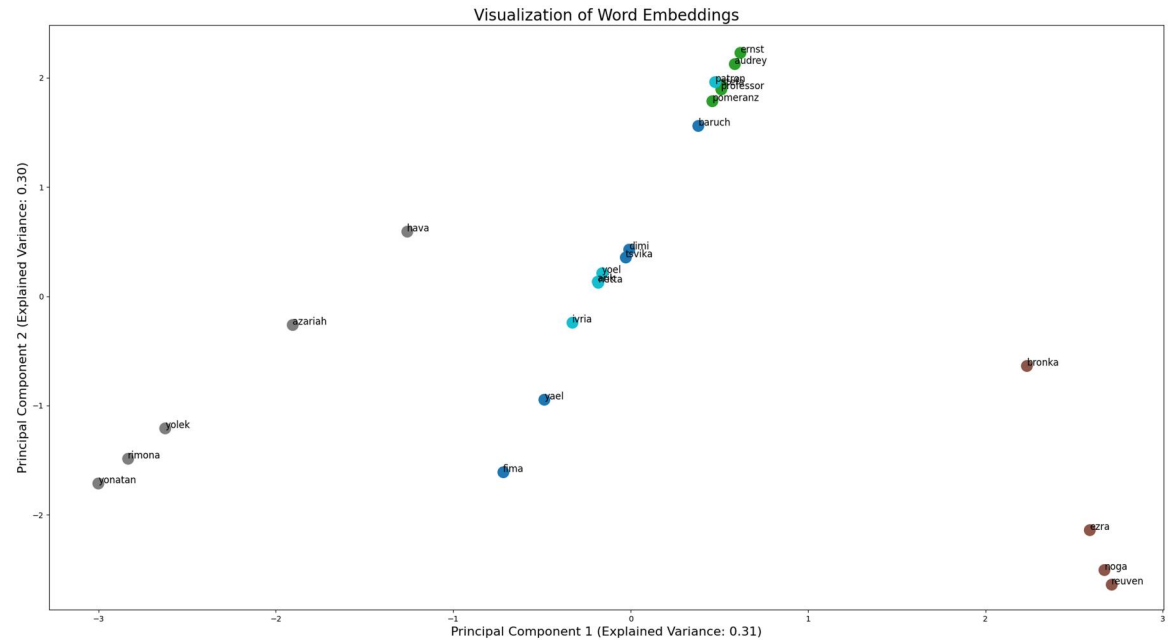


Figure 10. PCA projection of character embeddings to a 2D space (characters from the same book given the same colour).

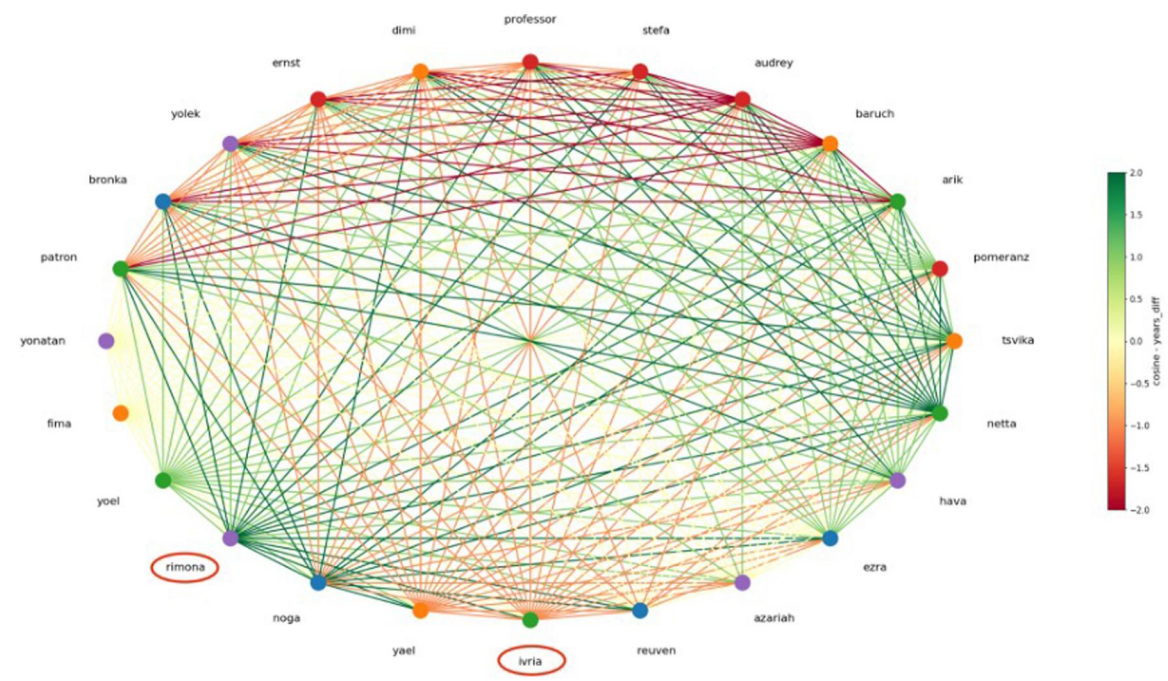


Figure 11. A gap network (cosine – 1/years_gap). ‘A Perfect Peace’ (Purple 1982), ‘Elsewhere Perhaps’ (Blue 1966), ‘Fima’ (Orange 1991), ‘To Know a Woman’ (Green 1989), ‘Touch the Water Touch the Wind’ (Red 1973).

the data: Nodes representing characters from the same book are given the same colour. This allows for easy identification of characters that belong to the same

literary work. The edges are coloured based on the absolute difference between the cluster numbers of the two graphs. The greener the edge, the higher the cosine

similarity compared to the years' gaps. Red marks the opposite direction. This colouring highlights significant deviations between the temporal and semantic relationships of the characters.

Again, these results are used as teasers for the researcher, who can decide what, if anything, to do with them. We can take a few thought-provoking examples from this rich information map. For instance, Rimona, the wife of Yonatan, the protagonist of *Perfect Peace* (1985)—and an extremely cold mother—has only green lines drawn from her (namely, semantic similarity overrides year difference). Indeed, there are good reasons to see her as an archetype character in Oz's work (it refers again to Oz's theme of motherhood). Her features (such as oddness) can be perceived in characters from different novels, regardless of their time of writing. An almost opposite example is Ivria, a main character in the novel 'To Know a Woman' (1989) discussed above, and Yael, a secondary character in *Fima* (1991) where almost all the lines drawn from them are orange. In that sense, they are unique.

7. Conclusions

Trilcke *et al.* (2017) showed how, by refining existing literary network methods, new real poetic discoveries can be achieved. In their case, it was the transition from static to dynamic networks, which are more sensitive to plot development. Following them, we, too, propose adding new elements—such as semantic, multi-layered, and cross-book networks—and, most importantly, offer them as teasers for experimental human heuristics. In this paper, we have presented the theoretical foundations of the approach, its methodological decisions, and briefly demonstrated how it was implemented in a sample corpus. Even if this approach does not fully answer the challenge presented at the beginning of the article, following Bakhtin's principle of dialogue, it nevertheless advances the research towards this goal. It should be noted, however, that the same approach can be applied to more common literary networks as well; every literary network, regardless of the method on which it is based, can be considered as a teaser for rereading.

It is theoretically possible to extend our basic idea. It is possible to add more and more layers to the integrated figures above, based on different computational methods, as well as different manual ones, and then give the results to the traditional researcher as a kind of 'textual MRI'. It opens up a world of endless interpretive possibilities, solid or thought-provoking. However, and what is more important than every visualization, this is not the end of the story. Under the framework of the back-and-forth interpretive

paradigm suggested here, every point of interest continues to be discussed and evaluated iteratively, integrating human reading and computational analysis.

The main question, for us, however, remains how to describe and frame it, or the 'so what' question. We believe that reframing the computational product as suggested above seems particularly important and appropriate for the current stage of computational literary studies. The field has passed beyond the stage of big conceptual promises that weren't always realized, and the so-called critical turn of it has already contributed to more balanced results. At the moment, however, given the tremendous progress of large language models (which can also be used for literary network analysis), there appears to be a risk that overconfidence in technology may undermine the fruits of such progress. Specifically, while large language models can tend to highlight the generic and the expected, the gaps-and-teasers approach suggested here seeks to highlight—to the extent possible—precisely the surprising and thought-provoking. In this regard, though focused on social literary networks, our proposal should be understood more broadly as an invitation to maintain a tension between distant and close readings—two meaning-making strategies whose connections, even now, are anything but trivial.

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Notes

1. Not surprisingly, Woloch's book seems to have a latent pre-computational dimension, which also had an effect on later computational works (such as Moretti 2013; Sack 2013).
2. The question goes beyond the boundaries of literary social networks; in fact, it applies to any attempt to turn a literary work into a computational model [CITE Marienberg-Milikowsky and Münz-Manor 2023]. But social literary networks seem a particularly fruitful test case for this, due to the gap between their intuitive simplicity, and the phenomena that they aim to address.
3. For a broader discussion of this vision, see Flüh *et al.* (2021).
4. See <https://dracor.org> (29 May 2024).
5. For a brief summary of these (and others) common methods, see Skorinkin (2017).
6. This seems to be a characteristic difference between NLP/CS and CLS/DH researchers.

7. For a summary and a demonstration of reader response theory, in all its various manifestations, see [Tompkins \(1980\)](#). The challenge that reader response theory posed for computational literary analysis was recognised as early as the seventies ([Wittig 1977](#)).
8. See also 'The Lives of Literary Characters' project, run by Andrew Piper and others: HYPERLINK "https://protect.checkpoint.com/v2/https://www.zooniverse.org/projects/citizen-readers/the-lives-of-literary-characters_.YzJlOmJlbmd1cmlybnVuaXZlcnNpdHlvZnRoZW5lZ2V2OmM6bzo5MjBhZTk1ZWJjNTVmYWwYjZkODhkZDYxYmU4YjRjZjo2Ojg3NTk6YjFmNzk4YjJmOWU4Njc2M2Q3MTY3ZWYwZDM3N2EzMDQxZGYzYmQ3ZTk1NmUyMjJlXOWU5MDJmMmQzMzZkODBiNzpwOjQ6Tg" <https://www.zooniverse.org/projects/citizen-readers/the-lives-of-literary-characters> (30 May 2024).
9. Thanks to an ongoing cooperation with the Hebrew Literature archives at the Hecksherim institute for Literature and Culture, Ben-Gurion University of the Negev.
10. The full project includes also an examination of source vs. translation computational analysis, which will not be described here. However, since names are one of the most easily translatable elements in literature, the effect of using books' translations appears to be limited. For a discussion of computational analysis of translations, see [Popovic et al. \(2023\)](#).
11. This web-based application is available here: HYPERLINK "https://protect.checkpoint.com/v2/https://ezlinavis.dracor.org_.YzJlOmJlbmd1cmlybnVuaXZlcnNpdHlvZnRoZW5lZ2V2OmM6bzo5MjBhZTk1ZWJjNTVmYWwYjZkODhkZDYxYmU4YjRjZjo2Ojg3MTY3ZWYwZDM3N2EzMDQxZGYzYmQ3ZTk1NmUyMjJlXOWU5MDJmMmQzMzZkODBiNzpwOjQ6Tg" <https://ezlinavis.dracor.org>.
12. Despite their shortcomings, we believe that accessible tools, for which a low skill threshold is required, are essential for integrating traditional researchers into computational research.
13. Moretti's aim was to better understand not the entire network as such, but the place of the protagonist as depicted by it, against another model, which has nothing to do with networks: 'Two distinct criteria for protagonism emerge: the volume of words, and the number of interactions ... the number of links tells us how connected a character is ...; the number of words tells us how much meaning the character brings into the play', while, of course, 'there are cases in which the two criteria of centrality coincide'.
14. Conceptually, our approach here follows the principled approach of Jan Christoph Meister, as described by [Flüh et al. \(2021\)](#). While Meister applied his approach to text annotation, we apply it to text network modelling.

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